Arrays in C

Arrays in C are a kind of data structure that can store a fixed-size sequential collection of elements of the same data type. Arrays are used to store a collection of data, but it is often more useful to think of an array as a collection of variables of the same type.

What is an Array in C?

An array in C is a collection of data items of similar data type. One or more values same data type, which may be primary data types (int, float, char), or user-defined types such as struct or pointers can be stored in an array. In C, the type of elements in the array should match with the data type of the array itself.

The size of the array, also called the length of the array, must be specified in the declaration itself. Once declared, the size of a C array cannot be changed. When an array is declared, the compiler allocates a continuous block of memory required to store the declared number of elements.

Why Do We Use Arrays in C?

Arrays are used to store and manipulate the similar type of data.

Suppose we want to store the marks of 10 students and find the average. We declare 10 different variables to store 10 different values as follows –

```
int a = 50, b = 55, c = 67, . . . ;
float avg = (float)(a + b + c +. . . ) / 10;
```

These variables will be scattered in the memory with no relation between them. Importantly, if we want to extend the problem of finding the average of 100 (or more) students, then it becomes impractical to declare so many individual variables.

Arrays offer a compact and memory-efficient solution. Since the elements in an array are stored in adjacent locations, we can easily access any element in relation to the current element. As each element has an index, it can be directly manipulated.

Example: Use of an Array in C

To go back to the problem of storing the marks of 10 students and find the average, the solution with the use of array would be –

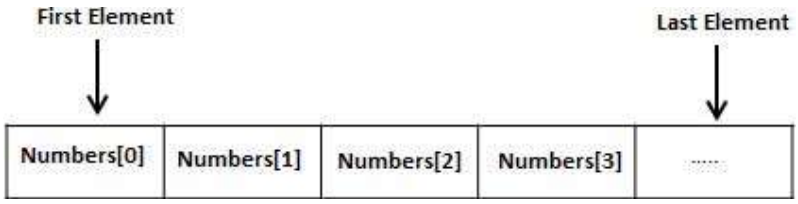
```c
#include <stdio.h>

int main(){
   int marks[10] = {50, 55, 67, 73, 45, 21, 39, 70, 49, 51};
   int i, sum = 0;
   float avg;

   for (i = 0; i <= 9; i++){
      sum += marks[i];
   }

   avg = (float)sum / 10;
   printf("Average: %f", avg);
   return 0;
}
```

Output

Run the code and check its output –

```
Average: 52.000000
```

Array elements are stored in contiguous memory locations. Each element is identified by an index starting with "0". The lowest address corresponds to the first element and the highest address to the last element.



Declaration of an Array in C

To declare an array in C, you need to specify the type of the elements and the number of elements to be stored in it.

Syntax to Declare an Array

```
type arrayName[size];
```

The "size" must be an integer constant greater than zero and its "type" can be any valid C data type. There are different ways in which an array is declared in C.

Example: Declaring an Array in C

In the following example, we are declaring an array of 5 integers and printing the indexes and values of all array elements –

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```
#include <stdio.h>

int main(){
   int arr[5];
   int i;

   for (i = 0; i <= 4; i++){
      printf("a[%d]: %d\n", i, arr[i]);
   }
   return 0;
}
```

Output

Run the code and check its output –

```
a[0]: -133071639
a[1]: 32767
a[2]: 100
a[3]: 0
a[4]: 4096
```

Initialization of an Array in C

At the time of declaring an array, you can initialize it by providing the set of comma-separated values enclosed within the curly braces {}.

Syntax to Initialize an Array

```
data_type array_name [size] = {value1, value2, value3, ...};
```

Example to Initialize an Array

The following example demonstrates the initialization of an integer array:

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```
// Initialization of an integer array
#include <stdio.h>

int main()
{
  int numbers[5] = {10, 20, 30, 40, 50};

  int i;  // loop counter

  // Printing array elements
  printf("The array elements are : ");
  for (i = 0; i < 5; i++) {
    printf("%d ", numbers[i]);
  }

  return 0;
}
```

Output

```
The array elements are : 10 20 30 40 50
```

Example of Initializing all Array Elements to 0

To initialize all elements to 0, put it inside curly brackets

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```
#include <stdio.h>

int main(){
```

```
   int arr[5] = {0};
   int i;

   for(i = 0; i <= 4; i++){
      printf("a[%d]: %d\n", i, arr[i]);
   }
   return 0;
}
```

Output

When you run this code, it will produce the following output –

```
a[0]: 0
a[1]: 0
a[2]: 0
a[3]: 0
a[4]: 0
```

Example of Partial Initialization of an Array

If the list of values is less than the size of the array, the rest of the elements are initialized with "0".

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```
#include <stdio.h>

int main(){
   int arr[5] = {1,2};
   int i;

   for(i = 0; i <= 4; i++){
      printf("a[%d]: %d\n", i, arr[i]);
   }
   return 0;
}
```

Output

When you run this code, it will produce the following output –

```
a[0]: 1
a[1]: 2
a[2]: 0
a[3]: 0
a[4]: 0
```

Example of Partial and Specific Elements Initialization

If an array is partially initialized, you can specify the element in the square brackets.

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```c
#include <stdio.h>

int main(){
   int a[5] = {1,2, [4] = 4};
   int i;

   for(i = 0; i <= 4; i++){
      printf("a[%d]: %d\n", i, a[i]);
   }
   return 0;
}
```

Output

On execution, it will produce the following output –

```
a[0]: 1
a[1]: 2
a[2]: 0
a[3]: 0
a[4]: 4
```

Getting Size of an Array in C

The compiler allocates a continuous block of memory. The size of the allocated memory depends on the data type of the array.

Example 1: Size of Integer Array

If an integer array of 5 elements is declared, the array size in number of bytes would be "sizeof(int) x 5"

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```
#include <stdio.h>

int main(){
   int arr[5] = {1, 2, 3, 4, 5};
   printf("Size of array: %ld", sizeof(arr));
   return 0;
}
```

Output

On execution, you will get the following output –

```
Size of array: 20
```

The **sizeof** operator returns the number of bytes occupied by the variable.

Example 2: Adjacent Address of Array Elements

The size of each **int** is 4 bytes. The compiler allocates adjacent locations to each element.

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```
#include <stdio.h>

int main(){
   int a[] = {1, 2, 3, 4, 5};
   int i;

   for(i = 0; i < 4; i++){
      printf("a[%d]: %d \t Address: %d\n", i, a[i], &a[i]);
   }
   return 0;
}
```

Output

Run the code and check its output –

```
a[0]: 1     Address: 2102703872
a[1]: 2     Address: 2102703876
a[2]: 3     Address: 2102703880
a[3]: 4     Address: 2102703884
```

In this array, each element is of **int** type. Hence, the 0th element occupies the first 4 bytes 642016 to 19. The element at the next subscript occupies the next 4 bytes and so on.

Example 3: Array of Double Type

If we have the array type of **double** type, then the element at each subscript occupies 8 bytes

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```
#include <stdio.h>

int main(){
   double a[] = {1.1, 2.2, 3.3, 4.4, 5.5};
   int i;

   for(i = 0; i < 4; i++){
      printf("a[%d]: %f \t Address: %ld\n", i, a[i], &a[i]);
   }
   return 0;
}
```

Output

Run the code and check its output –

```
a[0]: 1.100000   Address: 140720746288624
a[1]: 2.200000   Address: 140720746288632
a[2]: 3.300000   Address: 140720746288640
a[3]: 4.400000   Address: 140720746288648
```

Example 4: Size of Character Array

The length of a "char" variable is 1 byte. Hence, a char array length will be equal to the array size.

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```
#include <stdio.h>

int  main(){
   char a[] = "Hello";
   int i;

   for (i=0; i<5; i++){
      printf("a[%d]: %c address: %ld\n", i, a[i], &a[i]);
   }
   return 0;
}
```

Output

Run the code and check its output –

```
a[0]: H address: 6422038
a[1]: e address: 6422039
a[2]: l address: 6422040
a[3]: l address: 6422041
a[4]: o address: 6422042
```

Accessing Array Elements in C

Each element in an array is identified by a unique incrementing index, stating with "0". To access the element by its index, this is done by placing the index of the element within square brackets after the name of the array.

The elements of an array are accessed by specifying the index (offset) of the desired element within the square brackets after the array name. For example –

```
double salary = balance[9];
```

The above statement will take the 10th element from the array and assign the value to the "salary".

Example to Access Array Elements in C

The following example shows how to use all the three above-mentioned concepts viz. declaration, assignment, and accessing arrays.

</> Open Compiler

```
#include <stdio.h>

int main(){

   int n[5];  /* n is an array of 5 integers */
   int i, j;

   /* initialize elements of array n to 0 */
   for(i = 0; i < 5; i++){
      n[i] = i + 100;
   }

   /* output each array element's value */
   for(j = 0; j < 5; j++){
      printf("n[%d] = %d\n", j, n[j]);
   }
   return 0;
}
```

Output

On running this code, you will get the following output –

```
n[0] = 100
n[1] = 101
n[2] = 102
n[3] = 103
n[4] = 104
```

The index gives random access to the array elements. An array may consist of struct variables, pointers and even other arrays as its elements.

More on C Arrays